Level - 3

Passage 1...

According to VSEPR model, molecules adopt geometries in which their valence electron pairs position themselves as far from each other as possible. The VSEPR model considers double and triple bonds to have slightly greater repulsive effects than single bonds because of the repulsive effect of π-electrons. However the lone pair creates the maximum repulsive effect.

1. Which of the following statement is false ?
   (a) SbF_4^- and SF_4 are isostructural
   (b) In IOF_5 the hybridization of central atom is sp^3d^2
   (c) Double bond(s) in SOF_4 and XeO_3F_2, is/are occupying equatorial position(s) of their respective geometry
   (d) None of these
2. Which of the following does not represent the isostructural pair?
   (a) SF_5^- and IF_5 (b) ClO_2F_3 and SOF_4 (c) SeF_3^+ and XeO_3 (d) None
3. Select the incorrect statement with respect to SO_2Cl_2 molecule :
   (a) It gives H_2SO_4 and HCl on hydrolysis at room temperature
   (b) It has two $d\pi$-$p\pi$ bonds between S and O bonded atoms
   (c) It is a polar molecule
   (d) None

Passage 2...

According to VBT any covalent bond will be formed by overlapping of atomic orbitals of bonded atoms provided atomic orbitals must be half-filled and electrons be in opposite spin. According to type of overlapping covalent bonds can be classified as (a) σ-bond (b) π-bond (c) δ -bond:

1. Which of the following set of orbitals does not produce nodal plane in xz-plane ?
   (a) $d_{yz} + d_{yz}$ (b) $d_{xy} + d_{xy}$ (c) $p_y + d_{xy}$ (d) None of these
2. The combination of orbital that can not produce non-bonding molecular orbital is (inter- nuclear axis is z-axis):
   (a) $p_y + d_{x^2-y^2}$ (b) $p_z + d_{yz}$ (c) $s + d_{xz}$ (d) $d_{xy} + d_{xy}$
3. If $F_2C_1 = C_2$ part of $F_2C_1 = C_2 = C_3 = C_4F_2$ lies in yz-plane, then incorrect statement is :
   (a) Nodal plane of π-bond between C_1 and C_2 lies in yz-plane, formed by sideways overlapping of p_x-orbitals
   (b) Nodal plane of π-bond between C_2 and C_3 lies in xz-plane, formed by sideways overlapping of p_y-orbitals
   (c) Nodal plane of π-bond between C_3 and C_4 lies in yz-plane, formed by sideways overlapping of p_y-orbitals
   (d) Nodal plane of π-bond between C_2 and C_3 lies in xy-plane, formed by sideways overlapping of p_z-orbitals

Passage 3...

If the central atom is of third row or below this in the periodic table, then lone pair will occupy a stereochemically inactive s-orbital and bonding will be through almost pure p-orbitals and bond angles are nearly 90°, if the substituent's electronegativity value is $\leq$ 2.5.

1. In which of the following option, covalent bond is having maximum s% character?
   (a) S — H bond in H_2S
   (b) P — H bond in PH_3
   (c) N — H bond in NH_3
   (d) All have equal s% character
2. Select incorrect statement regarding P_4 molecule.
   (a) Each P atom is joined with three P-atoms
   (b) P_4 molecule contains total 12 bond angles
   (c) Lone pair of each P atom is present in pure s-orbital
   (d) Lone pair of each P atom present in hybrid orbital
3. The hybridisation of atomic orbitals of central atom "Xe" in XeO_4, XeO_2F_2 and $XeOF_4$ respectively.
   (a) sp^3, sp^3d^2, sp^3d^2
   (b) sp^3d, sp^3d, sp^3d^2
   (c) sp^3, sp^3d^2, sp^3d
   (d) sp^3, sp^3d, sp^3d^2

Passage 4...

According to V.B.T., atoms of element form bond only to pair up their unpaired electrons present in ground state or excited state. This pairing of unpaired electron will take place by overlapping of orbitals each one having one unpaired electron with opposite spin.

1. Which of the following orbital combination does not form π-bond?
   (a) $p_x + p_x$ sideways overlapping
   (b) $d_{x^2-y^2} + p_y$ sideways overlapping
   (c) $d_{xy} + d_{xy}$ sideways overlapping
   (d) $d_{yz} + p_y$ sideways overlapping
2. Which of the following orbital cannot form δ-bond?
   (a) $d_{x^2-y^2}$ orbital
   (b) d_{xy} orbital
   (c) d_{z^2} orbital
   (d) d_{zx} orbital
3. Which of the following combination of orbitals does not from any type of covalent bond (if z-axis is molecular axis)?
   (a) $p_z + p_z$
   (b) $p_y + p_y$
   (c) $s + p_y$
   (d) $s + s$

Passage 5...

The space model which is obtained by joining the points representing various bonded atoms gives the shape of the molecule. The geometry of the molecule is definite relative arrangement of the bonded atoms in a molecule. The shape and geometry of a molecule is explained by valence shell electron pair repulsion theory given by Gillespie and Nyholm.

1. Select the correct code for the following repulsion orders, according to VSEPR theory :
   (I) lone pair–lone pair > lone pair–bond pair
   (II) lone pair–bond pair > bond pair–bond pair
   (III) lone pair–lone pair > bond pair–bond pair

(IV) lone pair–bond pair > lone pair–lone pair

(a) I, II & III (b) II & IV (c) I, II & IV (d) All

2. Which molecule has both shape and geometry identical?

(I) $SnCl_2$ (II) NH_3 (III) PCl_5 (IV) SF_6

(a) I, III & IV (b) II, III & IV

(c) III & IV (d) All

3. Which is not the electron geometry of covalent molecules?

(a) Pentagonal bipyramidal (b) Octahedral

(c) Hexagonal (d) Tetrahedral

Passage 6...

When hybridisation involving d-orbitals are considered then all the five d-orbitals are not degenerate, rather $d_{x^2-y^2}$, d_{z^2} and d_{xy}, d_{yz}, d_{zx} form two different sets of orbitals and orbitals of appropriate set is involved in the hybridisation.

1. In sp^3d^2 hybridisation, which sets of d-orbitals is involved ?

(a) $d_{x^2-y^2}, d_{z^2}$ (b) d_{z^2}, d_{xy} (c) d_{xy}, d_{yz} (d) $d_{x^2-y^2}, d_{xy}$

2. In sp^3d^3 hybridisation, which orbitals are involved?

(a) $d_{x^2-y^2}, d_{z^2}, d_{xy}$ (b) d_{xy}, d_{yz}, d_{zx}

(c) $d_{x^2-y^2}, d_{xy}, d_{xz}$ (d) d_{z^2}, d_{yz}, d_{zx}

3. Molecule having trigonal bipyramidal geometry and sp^3d hybridisation, d-orbitals involved is:

(a) d_{xy} (b) d_{yz} (c) $d_{x^2-y^2}$ (d) d_{z^2}

4. Which of the following orbitals can not undergo hybridisation amongst themselves.

(I) $3d, 4s$ (II) $3d, 4d$

(III) $3d, 4s$ & $4p$ (IV) $3s, 3p$ & $4s$

(a) only II (b) II & III (c) I, II & IV (d) II & IV

Passage 7...

Ionic bond is defined as the electrostatic force of attraction holding the oppositely charged ions. Ionic compounds are mostly crystalline solid having high melting and boiling points, electrical conductivity in molten state, solubility in water etc. Covalent bond is defined as the force which binds atoms of same or different elements by mutual sharing of electrons in a covalent bond. Covalent compounds are solids, liquids or gases. They have low melting and boiling points compounds. They are more soluble in non-polar solvents.

1. The valence electrons are involved in formation of covalent bonds is/are called :

(a) non-bonding electrons (b) lone pairs

(c) unshared pairs (d) none of these

2. The amount of energy released when one mole of ionic solid is formed by packing of gaseous ion is called :

(a) Ionisation energy (b) Solvation energy

(c) Lattice energy (d) Hydration energy

3. Which of the following is arranged order of increasing boiling point ?

(a) $H_2O < CCl_4 < CS_2 < CO_2$ (b) $CO_2 < CS_2 < CCl_4 < H_2O$

(c) $CS_2 < H_2O < CO_2 < CCl_4$ (d) $CCl_4 < H_2O < CO_2 < CS_2$

Passage 8...

When an ionic compound is dissolved in water (polar solvent), it breaks up into its constituent ions. The given ionic compound will be dissolved in water if its hydration energy is more than lattice energy. If hydration energy is less than lattice energy then ionic compound is usually either sparingly soluble or insoluble in water.

1. Which of the following ionic compound is having maximum lattice energy :
   (a) NaF (b) MgF_2 (c) AlF_3 (d) KF
2. Most hydrated cation is :
   (a) $Ce^{4+}_{(aq.)}$ (b) $La^{3+}_{(aq.)}$ (c) $Ba^{2+}_{(aq.)}$ (d) $Cs^{+}_{(aq.)}$

Passage 9...

A covalent bond will be formed by the overlapping of atomic orbitals having single electron of opposite spin, according to the overlapping of atomic orbitals the covalent bond may be of two types :

(i) Sigma bond (ii) Pi bond (π)

Sigma bond is stronger bond than the Pi-bond. If atomic orbitals overlap about the nuclear axis then sigma bond is formed but when atomic orbitals overlap sideway then Pi-bond is formed.

1. The correct order of increasing C — O bond length of CO, CO_3^{2-}, CO_2 is :
   (a) $CO_3^{2-} < CO_2 < CO$ (b) $CO_2 < CO_3^{2-} < CO$
   (c) $CO < CO_3^{2-} < CO_2$ (d) $CO < CO_2 < CO_3^{2-}$
2. Compound having maximum bond angle is :
   (a) BBr_3 (b) BCl_3
   (c) BF_3 (d) None of these
3. The strength of bonds formed by $2s-2s$, $2p-2p$ and $2p-2s$ overlap has the order :
   (a) $s-s > p-p > p-s$ (b) $s-s > p-s > p-p$
   (c) $p-p > p-s > s-s$ (d) $p-p > s-s > p-s$

Passage 10...

According to VBT the extent of overlapping depends upon types of orbitals involved in overlapping and nature of overlapping. More will be the overlapping and the bond energy will also be high.

1. The incorrect order of bond dissociation energy will be :
   (a) $H-H > Cl-Cl > Br-Br$ (b) $Si-Si > P-P > Cl-Cl$
   (c) $C-C > N-N > O-O$ (d) $H-Cl > H-Br > H-I$
2. Which of the following combination of orbitals does not form covalent bond (x-axis is inter nuclear axis) :
   (a) $s + p_y$ (b) $p_y + p_y$
   (c) $d_{yz} + d_{yz}$ (d) $d_{xy} + d_{xy}$
3. Which of the following compound does not form $p\pi - p\pi$ bond ?
   (a) SO_3 (b) NO_3^-
   (c) SO_4^{2-} (d) CO_3^{2-}

Passage 11...

Consider the following elements with their period number and valence electrons.

Elements	Period number	Total valence e^-
P	2	4
Q	2	6
R	3	7
S	3	3
T	3	6
U	3	4

According to the given informations, answer the following questions :

1. Choose incorrect statement :
   (a) *R* exhibits maximum covalency among all elements given
   (b) *Q* does not exhibit variable covalency
   (c) *R* exhibits minimum covalency among all elements given
   (d) *R* and *S* combine each other and form SR_5 type of compound
2. Choose the correct statement :
   (a) *Q* has maximum value of electron affinity
   (b) *R* has maximum value of electronegativity
   (c) *S* has maximum atomic size
   (d) *T* and *U* are same group elements
3. Choose the incorrect statement :
   (a) SR_3 is a hypovalent compound
   (b) UR_4 can act as a Lewis acid
   (c) PQ_2 can not acts as Lewis acid
   (d) $UR_4 > SR_3$: Lewis acidic character

Passage 12...

Hybridisation involves the mixing of orbitals having comparable energies of same atom. Hybridised orbitals perform efficient overlapping than overlapping by pure *s*, *p* or *d* orbitals.

1. Which of the following is not correctly match between given species and type of overlapping ?
   (a) XeO_3: Three $(d\pi - p\pi)$ bonds
   (b) H_2SO_4: Two $(d\pi - p\pi)$ bonds
   (c) SO_3: Three $(d\pi - p\pi)$ bonds
   (d) $HClO_4$: Three $(d\pi - p\pi)$ bonds
2. Consider the following compounds and select the incorrect statement from the following :

$$NH_3,\ PH_3,\ H_2S,\ SO_2,\ SO_3,\ BF_3,\ PCl_3,\ IF_7,\ P_4,\ H_2$$

   (a) Six molecules out of given compounds involves hybridisation
   (b) Three molecules are hypervalent compounds
   (c) Six molecules out of above compounds are non-planar in structure
   (d) Two molecules out of given compounds involves $(d\pi - p\pi)$ bonding as well as also involves $(p\pi - p\pi)$ bonding

Passage 13...

Lattice energy is the amount of energy release when one mole of ionic solid is formed from its constitutent gaseous cations and anions.
Hydration energy is the amount of energy release when one mole of gaseous cation or one mole of gaseous anion is completely hydrated.

1. Consider the following conversion and calculate the net energy change for the following conversion :

$$NaCl(s) + H_2O \longrightarrow Na^+(aq) + Cl^-(aq)$$

Given : LE of NaCl $= x$ kJ/mol

HE of Na^+ $= y$ kJ/mol

HE of Cl^- $= z$ kJ/mol

(a) $(x - y + z)$ kJ/mol
(b) $(-x + y + z)$ kJ/mol
(c) $(x - y - z)$ kJ/mol
(d) $(x + y + z)$ kJ/mol

2. Select the correct order :

(a) $Na_3N > Mg_3N_2 > AlN$ (Lattice energy)

(b) $Na^+(g) < Mg^{2+}(g) < Al^{3+}(g)$ (Ionic radius)

(c) $Li^+(aq) > Na^+(aq) > K^+(aq)$ (Hydrated ion radius)

(d) $F^-(aq) > Cl^-(aq) > I^-(aq)$ (Ionic mobility)

Passage 14...

It is cleared from the Kossel and Lewis approach that the formation of an ionic compound would primarily depends upon the ease of formation of the positive and negative ions from the respective neutral atoms. Conditions for the formation of ionic compounds are :

(i) Electronegativity difference between two combining elements must be larger.

(ii) Ionization enthalpy $[(M(g) \rightarrow M^+(g) + e^-)]$ of electropositive element must be low.

(iii) Negative value of electron gain enthalpy $[(X(g) + e^- \rightarrow X^-(g))]$ of electronegative element should be high.

(iv) Lattice enthalpy $[(M^+(g) + X^-(g) \rightarrow MX(s))]$ of an ionic solid must be high.

1. Select the correct reason for given statement "Flouride of Al is ionic while chloride of Al is covalent".

(a) IE of Al in $AlF_3 > AlCl_3$
(b) EA of Flourine > Chlorine
(c) Lattice energy of $AlF_3 > AlCl_3$
(d) EA of Flourine < Chlorine

2. Which of the following is ionic compound ?

(a) $(NH_4)_2SO_4$
(b) HCN
(c) AlI_3
(d) HBr

Passage 15...

ACl_3 is trigonal pyramidal and DCl_3 is trigonal planar compound, when these two compounds ACl_3 and DCl_3 are mixed together then a compound $ADCl_6$ is formed. Structural analysis showed that $ADCl_6$ is an ionic compound, now answer the following with respect to given conditions.

1. If anion has see-saw shape then shape of cation formed is :

(a) linear
(b) bent
(c) pentagonal bipyramidal
(d) trigonal planar

2. If shape of anion is tetrahedral then shape of cation formed is :

(a) tetrahedral
(b) bent
(c) linear
(d) trigonal planar

One or More Answers is/are Correct

1. In which of the following there is intermolecular hydrogen bonding?
(a) Water
(b) Ethanol
(c) Acetic acid
(d) H—F

2. Correct order of decreasing boiling points is :
(a) $HF > HI > HBr > HCl$
(b) $H_2O > H_2Te > H_2Se > H_2S$
(c) $Br_2 > Cl_2 > F_2$
(d) $CH_4 > GeH_4 > SiH_4$

3. In which species the hybrid state of central atom is/are sp^3d ?
(a) I_3^-
(b) SF_4
(c) PF_5
(d) IF_5

4. Select correct statement(s) is/are :
(a) In AsH_3 molecule lone pair at central atom is present in almost pure *s*-orbital
(b) Number of $p\pi - d\pi$ bond in SO_3 and SO_2 are same
(c) NF_3 is better Lewis base than NCl_3
(d) Stable oxidation state of Lead is +2

5. Which of the following species does/do not exist?
(a) OF_4
(b) NH_2^-
(c) NCl_5
(d) ICl_3^{2-}

6. Which of the following species is/are superoctet molecule?
(a) AlF_3
(b) $SiCl_4$
(c) XeF_2
(d) ICl_3

7. Which of the following statements is incorrect ?
(a) A σ-bond is weaker than a π-bond
(b) There are four co-ordinate bonds in the NH_4^+ ions
(c) The covalent bond is directional in nature
(d) HF is less polar than HCl

8. Which of the following species is/are capable of forming a coordinate bond with BF_3?
(a) PH_3
(b) NH_4^+
(c) OH^-
(d) Mg^{2+}

9. Ionic compounds in general do not possess :
(a) high melting points and non-directional bonds
(b) high melting points and low-boiling points
(c) directional bonds and low-boiling points
(d) high solubilities in polar and non-polar solvents

10. Correct stability order of metal cation is/are :
(a) $Pb^{2+} < Sn^{2+}$
(b) $Pb^{4+} < Pb^{2+}$
(c) $Sn^{4+} < Sn^{2+}$
(d) $Pb^{4+} < Sn^{4+}$

11. Consider the following molecule :

$$\underset{(1)}{H_2C} = \underset{(2)}{C} = \underset{(3)}{C} = \underset{(4)}{C} = \underset{(5)}{CF_2}$$

If hybridization of $C_{(1)}$ carbon atom is $sp^2(s + p_y + p_z)$ and hybridization of $C_{(4)}$ carbon atom is $sp(s + p_z)$.
Then according to given information the **correct** statement(s) is/are :

(a) Nodal plane of π-bond between $C_{(2)}$ and $C_{(3)}$ lies in xz-plane, formed by sideways overlapping p_y-orbitals

(b) Nodal plane of π-bond between $C_{(3)}$ and $C_{(4)}$ lies in yz-plane, formed by side ways overlapping p_x-orbitals

(c) The orbitals involve in hybridization of $C_{(5)}$ carbon atom are $s + p_x + p_z$

(d) Nodal plane of π-bond between $C_{(1)}$ and $C_{(2)}$ lies in yz-plane, formed by side ways overlapping p_y-orbitals

12. Consider the following two molecules and according to the given information select correct statement about AX_2 and AY_2 :

where A : 16th group of 3rd period element

X : more electronegative than (A) and same group number of (A)

Y : Less atomic size than (A) and same period number of (A)

(a) The hybridization of central atoms are different in both compounds

(b) The shape of both molecules are same

(c) Both compounds are planar

(d) The $X - \hat{A} - X$ bond angle is less than $Y - \hat{A} - Y$ bond angle

13. Which of the following statements are correct about sulphur hexafluoride ?

(a) all S — F bonds are equivalent

(b) SF_6 is a planar molecule

(c) oxidation number of sulphur is the same as number of electrons of sulphur involved in bonding

(d) sulphur has acquired the electronic structure of the gas argon

14. If AB_4^n, types species are tetrahedral, then which of the following is/are correctly match? (Where A central atom, B is surrounding atom and n is charge on species.)

A	B	n
(a) Xe	O	0
(b) Se	F	0
(c) P	O	–3
(d) N	H	+1

15. Which of the following statements is correct ?

(a) ClF_3 molecule is bent T-shape

(b) In SF_4 molecule, F—S—F equatorial bond angle is 103° due to $lp - lp$ repulsion

(c) In $[ICl_4]^-$ molecular ion, Cl—I—Cl bond angle is 90°

(d) In OBr_2, the bond angle is less than OCl_2

16. Which of the following combination of bond pair (b.p.) and lone pair (l.p.) give same shape?

(i) 3 b.p.+1 l.p. (ii) 2 b.p.+2 l.p. (iii) 2 b.p.+1 l.p. (iv) 2 b.p.+0 l.p.

(v) 3 b.p.+2 l.p. (vi) 2 b.p.+3 l.p.

(a) (ii) and (iii)

(b) (iv) and (v)

(c) (iv) and (vi)

(d) (iii) and (vi)

17. Select the true statement(s) among the following :

(a) Pure overlapping of two d_{xy} orbitals along x-axis results in the formation of π-bond

(b) $NO_2^+ > NO_3^- > NO_2^-$ is the correct order of bond angle as well as N—O bond order

(c) $NF_3 < NCl_3 < NBr_3 < NI_3$ is the correct order of Lewis basic character as well as bond angle

(d) HF > HCl > HBr > HI is the correct order of dipole moment as well as boiling point

18. p_y-orbital can not form π-bond by lateral overlap with :

(a) d_{xz}-orbital (b) $d_{x^2-y^2}$-orbital

(c) d_{xy}-orbital (d) p_z-orbital

19. Which of the following orbital(s) cannot form δ-bond ?

(a) $d_{x^2-y^2}$-orbital (b) d_{xy}-orbital

(c) d_{z^2}-orbital (d) p_x-orbital

20. Select correct statement(s) regarding σ and π-bonds:

(a) σ-bond lies on the line joining the nuclei of bonded atoms

(b) π-electron cloud lies on either side to the line joining the nuclei of bonded atoms

(c) $(2p_\pi - 3d_\pi)$ π-bond is stronger than $(2p_\pi - 3p_\pi)$ π-bond.

(d) σ-bond has primary effect to decide direction of covalent bond, while π-bond has no primary effect in direction of bond

21. Which of the following statements is/are correct?

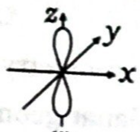
(a) All carbon to carbon bonds contain a sigma bond and one or more π-bonds

(b) All carbon to carbon bonds are sigma bonds

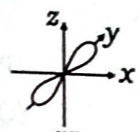
(c) All oxygen to hydrogen bonds are hydrogen bonds

(d) All carbon to hydrogen bonds are sigma bonds

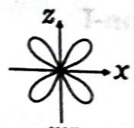
22. Consider the following three orbitals :



(i)



(ii)



(iii)

Correct statement(s) regarding given information is/are :

(a) Orbitals (i) and (ii) can never form any type of covalent bond

(b) If internuclear axis is 'x', then combination of (ii) and (iii) orbitals can form π-bond

(c) Orbital (iii) can form δ-bond with other orbital having same orientation of lobes

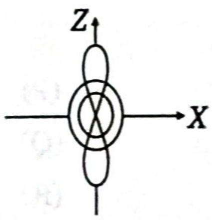
(d) If internuclear axis is 'x', then combination of (i) and (iii) orbitals can form π-bond

23. Which of the following combination of orbitals do/does not form bond (if x-axis is internuclear axis) ?

(a) $s + p_z$ (b) $s + s$

(c) $p_z + p_x$ (d) $d_{xy} + p_y$

24. Consider the following atomic orbitals :



Which of the following statement(s) is/are correct regarding given orbital ?

(a) It is a gerade atomic orbital

(b) It has zero nodal plane

(c) Circular electron density is present in 'XY' plane

(d) Opposite lobes of orbital have same sign of wave function (ψ)

25. Select correct order for $A(g)$ and $B(g)$ where salt $AB(s) = A^+B^-$ is most ionic :

(a) Ionization energy of $A < B$ (b) Electron affinity of $A < B$

(c) Atomic size of $A < B$ (d) Electro negativity of $A < B$

26. Which pair(s) can form 'XY' type compound ?
(a) Al, P (b) Mg, N (c) Ca, O (d) Na, F

27. Correct statement(s) regarding ionic compound is/are :
(a) High melting points and non-directional bonds
(b) Shows structural isomerism
(c) Directional bonds and low-boiling points
(d) High solubilities in polar solvents

28. Pick out among the following species isoelectronic with CO_2 :
(a) N_3^- (b) $(CNO)^-$ (c) $(NCN)^{2-}$ (d) NO_2^-

29. Which of the following molecular species does/do not exist ?
(a) F_3^- (b) SCl_6 (c) ICl_6^- (d) $PCl_6^-(s)$

Match the Column

Column-I and Column-II contains four entries each. Entries of Column-I are to be matched with some entries of Column-II. One or more than one entries of Column-I may have the matching with the same entries of Column-II.

1.

Column-I	Column-II
(A) $B_3N_3H_6$	(P) Planar geometry
(B) I_3^-	(Q) Non-planar geometry
(C) B_2Cl_4 (Solid)	(R) Compound having coordinate bond
(D) SiF_4	(S) Compound having back bond
	(T) Non-polar compound

2.

Column-I (Shape)	Column-II (Hybridisation)
(A) Linear	(P) sp^3
(B) Angular	(Q) sp^3d^2
(C) Square planar	(R) sp^2
(D) Trigonal planar	(S) sp^3d

3.

Column-I	Column-II
(A) SO_3	(P) Largest bond angle
(B) $BeCl_2$	(Q) Lowest bond angle
(C) NH_3	(R) sp^2-hybridisation
(D) NO_2^-	(S) sp^3-hybridisation

4.

Column-I	Column-II
(A) Hypo phosphoric acid	(P) All hydrogen are ionizable in water
(B) Pyro phosphorous acid	(Q) Lewis acid in water
(C) Boric acid	(R) Monobasic
(D) Hypo phosphorous acid	(S) sp^3-hybridised central atom

5.

Column-I	Column-II
(A) NH_2^-	(P) Square pyramidal
(B) $XeOF_2$	(Q) V-shaped
(C) ICl_4^-	(R) T-shaped
(D) $[SbF_5]^{2-}$	(S) Square planar

6.

Column-I	Column-II
(A) ICl_2^-	(P) Linear
(B) BrF_2^+	(Q) Pyramidal
(C) ClF_4^-	(R) Tetrahedral
(D) $AlCl_4^-$	(S) Square planar
	(T) Angular

7.

Column-I	Column-II
(A) $Re_2Cl_8^{2-}$	(P) $p\pi - p\pi$ bonding
(B) NO_3^-	(Q) $p\pi - d\pi$ bonding
(C) SO_4^{2-}	(R) $d\pi - d\pi$ bonding
(D) SO_3	(S) δ-bonding

8.

Column-I (Bond order range)	Column-II (Oxyanions)
(A) 1.0 to 1.30	(P) NO_3^-
(B) 1.31 to 1.55	(Q) ClO_4^-
(C) 1.56 to 1.70	(R) PO_4^{3-}
(D) 1.71 to 2.0	(S) ClO_3^-
	(T) SO_4^{2-}

9.

Column-I	Column-II
(A) AsO_4^{3-}	(P) All three *p*-orbitals used in hybridisation
(B) ICl_2^+	(Q) Tetrahedral shape
(C) SOF_4	(R) Axial *d*-orbital with two nodal cones used in hybridisation
(D) $XeOF_4$	(S) All bond lengths are identical
	(T) $p\pi - d\pi$ bond(s) present

10.

Column-I σ-bond pairs + lone pairs around central atom of AB_x type compound	Column-II Characteristics/shape of compound
(A) 2 + 1	(P) Linear
(B) 2 + 3	(Q) Angular
(C) 4 + 2	(R) Polar
(D) 2 + 2	(S) Non-polar
	(T) Planar

11.

Column-I (Type of bond formed)	Column-II [Combining orbitals (Internuclear axis)]
(A) π-bond	(P) $d_{yz} + p_y$, (z)
(B) σ-bond	(Q) $s + p_x$, (y)
(C) δ-bond	(R) $d_{yz} + d_{yz}$, (x)
(D) Non-bonding	(S) $s + s$, (z)
	(T) $s + d_{xy}$, (y)

12.

Column-I	Column-II
(A) XeF_5^-	(P) d-orbital with zero nodal plane is used in hybridisation
(B) PBr_4^+	(Q) Non-axial d-orbital is used in hybridisation
(C) IOF_3	(R) Planar species
(D) NH_2^-	(S) Non-planar species
	(T) Bond angle 109°28' or less than 109°28'

13.

Column-I	Column-II
(A) $IO_2F_2^-$	(P) Trigonal pyramidal shape
(B) IOF_4^-	(Q) Square pyramidal shape
(C) $SeOF_2$	(R) See-saw shape
(D) $XeOF_2$	(S) Non-planar
	(T) One of the bond angle < 90°

Assertion-Reason Type Questions

These questions consist of two statements each, printed as assertion and reason, while answering these questions you are required to choose any one of the following responses.

(A) If assertion is true but the reason is false

(B) If assertion is false but reason is true

(C) If both assertion and reason are true and the reason is the correct explanation of assertion

(D) If both assertion and reason are true but reason is not the correct explanation of assertion

1. **Assertion** : Multiple bond between two bonded atoms can have more than three bonds.
   Reason : Multiple bond between two bonded atoms can not have more than two π-bonds.
2. **Assertion** : 2^{nd} period elements do not involve in excitation of electron.
   Reason : 2^{nd} period elements do not have vacant 2d-orbitals.
3. **Assertion** : In SO_3 molecule bond dissociation energy of all S=O bonds are not equivalent.
   Reason : SO_3 molecule is having two types of $(2p\pi - 3p\pi)$ and $(2p\pi - 3d\pi)$ pi-bonds.
4. **Assertion** : PH_4^+ ion is having tetrahedron geometry.
   Reason : P-atom is unhybridised in PH_4^+ ion.
5. **Assertion** : All diatomic molecules with polar bond have dipole moment.
   Reason : Dipole moment is a vector quantity.

6. **Assertion** : Water is a good solvent for ionic compounds but poor for covalent compounds.
Reason : Hydration energy of ions releases sufficient energy to overcome lattice energy and break hydrogen bonds in water while covalent compounds interact so weakly that even van der Waals' forces between molecules of covalent compounds cannot be broken.
7. **Assertion** : Xe-atom in XeF_2 assumes *sp*-hybrid state.
Reason : XeF_2 molecule does not follow octet rule.
8. **Assertion** : The atoms in a covalent molecule are said to share electrons, yet some covalent molecules are polar.
Reason : In polar covalent molecule, the shared electrons spend more time on the average near one of the atoms.
9. **Assertion** : CCl_4 is a non-polar molecule.
Reason : CCl_4 has polar bonds.
10. **Assertion** : Geometry of ICl_3 is tetrahedral.
Reason : Its shape is T-shape, due to the presence of two lone pairs.
11. **Assertion** : The covalency of carbon is four in excited state.
Reason : The four half-filled pure orbitals of carbon form same kind of bonds with an atom as those are with hybridised orbitals.
12. **Assertion** : The shape of XeF_4 is square-planar.
Reason : In an octahedral geometry, a single lone pair can occupy any position but a second lone pair will occupy the opposite position to the first lone pair.

Subjective Problems

1. Consider following compounds A to E :
(A) XeF_n (B) $XeF^{+}_{(n+1)}$ (C) $XeF^{-}_{(n+1)}$ (D) $XeF_{(n+2)}$
(E) $XeF^{2-}_{(n+4)}$,
If value of n is 4, then calculate value of "$p \div q$" here, 'p' is total number of bond pair and 'q' is total number of lone pair on central atoms of compounds (A) to (E).
2. Consider the following species and find out total number of species which are polar and can act as Lewis acid
CCl_4, CO_2, SO_2, $AlCl_3$, HCHO, SO_3, $SiCl_4$, BCl_3, CF_4
3. Consider the following table regarding interhalogen compounds, XY_n (where Y is more electronegative than X)

Value of n for respective interhalogen compound	Total number of d-orbitals used in hybridization of central atom	Polarity	Planarity
P_1	1	Polar	Planar
P_2	Q_1	Polar	Non-Planar
P_3	Q_2	Non-Polar	Non-Planar

Then according to given information calculate value of expression $P_2 \times \frac{(P_3 - P_1)}{(Q_1 + Q_2)}$.
4. What is covalency of chlorine atom in second excited state?
5. Sum of σ and π-bonds in NH_4^+ cation is .....
6. Calculate the value of $X - Y$, for $XeOF_4$. (X = Number of σ bond pair and Y = Number of lone pair on central atom)

7. The molecule ML_x is planar with 6 electron pairs around M in the valence shell. The value of x is :

8. Calculate value of $\frac{X+Y+Z}{10}$, here X is O—N—O bond angle in NO_3^- Y is O—N—O bond angle in NO_2^+ and Z is F—Xe—F adjacent bond angle in XeF_4.

9. Calculate $x+y+z$ for H_3PO_3 acid, where x is no. of lone pairs, y is no. of σ bonds and z is no. of π bonds.

10. How many right angle, bond angles are present in TeF_5^- molecular ion?

11. How may possible ∠ FSeF bond angles are present in SeF_4 molecule?

12. In IF_6^- and TeF_5^-, sum of axial d-orbitals which are used in hybridisation in both species.

13. Among the following, total no. of planar species is:

(i) SF_4 (ii) BrF_3 (iii) XeF_2 (iv) IF_5
(v) SbF_4^- (vi) SF_5^- (vii) SeF_3^+ (viii) CH_3^+
(ix) PCl_4^+

14. Calculate the value of "$x+y-z$" here x, y and z are total number of non-bonded electron pair(s), pie(π) bond(s) and sigma(σ) bonds in hydrogen phosphite ion respectively.

15. Consider the following table

Total number of electron pairs (l.p. +σ-bond)	Total number of lone pairs	Shape
5	....p....	linear
....q....	1	see-saw
4	....r....	Bent shape
....s....	2	Square planar
5	....t....	Bent 'T' shape

Then calculate value of "$p+q+r-s-t$".

16. In phosphorus acid, if X is number of non bonding electron pairs. Y is number of σ-bonds and Z is number of π-bonds. Then, calculate value of "$Y \times Z - X$".

17. Calculate the number of $p_\pi - d_\pi$ bond(s) present in SO_4^{2-} :

18. Sum of σ & π-bonds in NH_4^+ cation is ....

19. Consider the following orbitals :

(i) $3p_x$ (ii) $4d_{z^2}$ (iii) $3d_{x^2-y^2}$ (iv) $3d_{yz}$

Then calculate value of '$x+y-z$' here x is total number of gerade orbital and y is total number of ungerade orbitals and z is total number of axial orbital in given above orbitals.

20. Calculate value of $|x-y|$, here, x and y are the total number of bonds in benzene and benzyne respectively which are formed by overlapping of hybridized orbitals.

21. Consider the following compounds :

(i) IF_5 (ii) ClI_4^- (iii) XeO_2F_2 (iv) NH_2^-
(v) BCl_3 (vi) $BeCl_2$ (vii) $AsCl_4^+$ (viii) $B(OH)_3$
(ix) NO_2^- (x) ClO_2^+

Then calculate value of "$x + y - z$", here, x, y and z are total number of compounds in given compounds in which central atom used their all three p-orbitals, only two p-orbitals and only one p-orbital in hybridisation respectively.

22. Total number of species which used all three p-orbitals in hybridisation of central atoms and should be non-polar also.

XeO_2F_2, $SnCl_2$, IF_5, I_3^+, XeO_4, SO_2, XeF_7^+, SeF_4

23. Consider the following species NO_3^-, SO_4^{2-}, ClO_3^-, SO_3, PO_4^{3-}, XeO_3, CO_3^{2-}, SO_3^{2-}

Then calculate value of $|x - y|$, where

x : Total number of species which have bond order 1.5 or greater than 1.5

y : Total number of species which have bond order less than 1.5

24. Consider the following orbitals $3s$, $2p_x$, $4d_{xy}$, $4d_{z^2}$, $3d_{x^2-y^2}$, $3p_y$, $4s$, $4p_z$ and find total number of orbital(s) having even number of nodal plane.

25. For the following molecules :

PCl_5, BrF_3, ICl_2^-, XeF_5^-, NO_3^-, XeO_2F_2, PCl_4^+, CH_3^+

Calculate the value of $\dfrac{a+b}{c}$

a = Number of species having sp^3d-hybridisation

b = Number of species which are planar

c = Number of species which are non-planar

26. Find out number of transformation among following which involves the change of hybridisation of underlined atom.

(a) $H_2\underline{O} + H^+ \longrightarrow H_3\underline{O}^+$

(b) $NH_3 + \underline{B}F_3 \longrightarrow NH_3 \cdot \underline{B}F_3$

(c) $\underline{Xe}F_6 \longrightarrow \underline{Xe}F_5^+ + F^-$

(d) $2\underline{P}Cl_5 \longrightarrow \underline{P}Cl_4^+ + PCl_6^-$

(e) $\underline{C}H_3 - CH_3 \longrightarrow \underline{C}H_3^- + CH_3^+$

27. Find total number of orbital which can overlap colaterally, (if inter nuclear axis is z)

$s, p_x, p_y, p_z, d_{xy}, d_{yz}, d_{xz}, d_{z^2}, d_{x^2-y^2}$

28. How many hydrogen bond(s) is/are formed by a molecule of H_3BO_3 in a layer of solid H_3BO_3?

29. Find out maximum number of bonds which lie in same plane in a molecule of IF_7.

30. Find out numerical value of expression $\left[\dfrac{x^2 + y^2}{2}\right]$ for $[NH_4]ICl_2$

where, x = Number of lone pair(s) on central atoms.

y = Maximum number of bond angles having same magnitude.

31. Find out numerical value of expression $\left|\dfrac{X+Y}{2}\right|$ for molecular species N_2 and C_2.

Where, X = Sum of numbers of filled molecular orbitals of N_2 and C_2 molecules which are formed by s-p mixing.

Y = Sum of numbers of filled molecular orbitals of N_2 and C_2 molecules which are not formed by s-p mixing.

32. Find out numerical value of expression $[a^2 \div 5]$ for molecular species : XeF_5^+, $XeOF_4$, $IO_2F_2^-$, IF_4^+, IOF_5, BrF_5, XeF_5^-, TeF_5^-, $[SiF_6]^{2-}$. Where; 'a' is number of molecular species in which central atom is sp^3d^2 hybridised as well as it contains one lone pair.

Level-3

Passage–1	1. (d)	2. (d)	3. (d)	
Passage–2	1. (d)	2. (d)	3. (c)	
Passage–3	1. (c)	2. (d)	3. (d)	
Passage–4	1. (b)	2. (c)	3. (c)	
Passage–5	1. (a)	2. (c)	3. (c)	
Passage–6	1. (a)	2. (a)	3. (d)	4. (d)
Passage–7	1. (d)	2. (c)	3. (b)	
Passage–8	1. (c)	2. (a)		
Passage–9	1. (d)	2. (d)	3. (c)	
Passage–10	1. (b)	2. (a)	3. (c)	
Passage–11	1. (d)	2. (c)	3. (c)	
Passage–12	1. (c)	2. (c)		
Passage–13	1. (c)	2. (c)		
Passage–14	1. (c)	2. (a)		
Passage–15	1. (a)	2. (b)		

One or More Answers is/are correct

1.	(a,b,c,d)	2.	(a,b,c)	3.	(a,b,c)	4.	(a,d)	5.	(a,c,d)	6.	(c,d)
7.	(a,b,d)	8.	(a,c)	9.	(b,c,d)	10.	(b,d)	11.	(a,b,c)	12.	(a,b,c)
13.	(a,c)	14.	(a,c,d)	15.	(a,c)	16.	(a,c)	17.	(a,c)	18.	(a,b,d)
19.	(c,d)	20.	(a,b,c,d)	21.	(d)	22.	(a,c,d)	23.	(a,c)	24.	(a,b,c,d)
25.	(a,b,d)	26.	(a,c,d)	27.	(a,d)	28.	(a,b,c)	29.	(a,b,c)		

Match the Column

1.	A→ P, S, T;	B→ P, R, T;	C→ P, S, T;	D→ Q, S, T
2.	A→ P, S;	B→ P, R;	C→ Q;	D→ R
3.	A→ R;	B→ P;	C→ Q, S;	D→ R
4.	A→ P, Q, S;	B→ Q, S;	C→ Q, R;	D→ Q, R, S
5.	A→ Q;	B→ R;	C→ S;	D→ P
6.	A→ P;	B→ T;	C→ S;	D→ R
7.	A→ R, S;	B→ P;	C→ Q;	D→ P, Q
8.	A→ R;	B→ P, T;	C→ S;	D→ Q
9.	A→ P, Q, S, T;	B→ P, S;	C→ P, R, T;	D→ P, R, T
10.	A→ Q, R, T;	B→ P, S, T;	C→ S, T;	D→ Q, R, T
11.	A→ P;	B→ S;	C→ R;	D→ Q, T
12.	A→ P, Q, R, T;	B→ S, T;	C→ P, S, T;	D→ R, T
13.	A→ R, S, T;	B→ Q, S, T;	C→ P, S;	D→ T

Assertion-Reason Type Questions

1.	(D)	2.	(B)	3.	(B)	4.	(A)	5.	(D)	6.	(C)	7.	(B)	8.	(C)	9.	(D)	10.	(B)
11.	(A)	12.	(C)																

Subjective Problems

1.	(4)	2.	(2)	3.	(4)	4.	(5)	5.	(4)	6.	(4)	7.	(4)	8.	(39)
9.	(13)	10.	(0)	11.	(6)	12.	(4)	13.	(3)	14.	(3)	15.	(2)	16.	(0)
17.	(2)	18.	(4)	19.	(1)	20.	(1)	21.	(8)	22.	(2)	23.	(0)	24.	(5)
25.	(3)	26.	(3)	27.	(6)	28.	(06.00)	29.	(05.00)	30.	(22.50)	31.	(06.50)	32.	(03.20)